

Ask Data - Detailed Project Guide

Generated: 2026-03-28 13:48:22

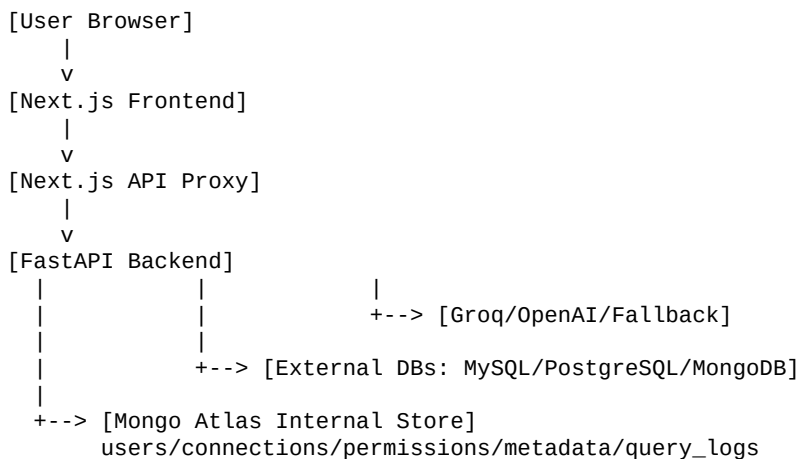
This version is a deep technical and operational walkthrough: workflows, API map, data model, tenant isolation, and file-by-file architecture notes.

1. Core Workflows

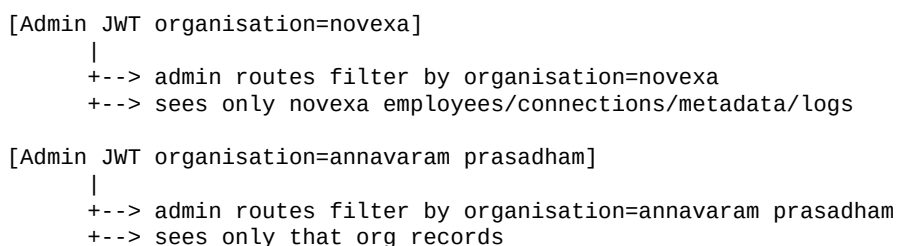
- Auth workflow: register -> login -> JWT -> role-based route access.
- Admin workflow: create employees, onboard connections, assign permissions.
- Metadata workflow: scan external DB schema and store in metadata_catalog.
- Employee workflow: query/analyse only assigned and organisation-scoped connections.
- Export workflow: convert analysis rows to csv/xlsx/pdf on demand.
- Isolation workflow: organisation filter enforced in admin and employee data paths.

2. Diagrams

System Architecture



Tenant Isolation



Connection Onboarding

```
Admin UI -> POST /admin/connections
-> sanitize payload -> encrypt secrets -> test connection
-> save with organisation -> refresh metadata
-> metadata_catalog updated
-> UI shows connection and schema rows
```

Employee Analysis

```
Employee UI -> POST /employee/analyse
-> permission check + org scope check
-> AI/fallback query generation
-> execute safe reads
-> rows + overview + analytics returned
```

-> optional export csv/excel/pdf

3. API Endpoint Map

Method	Route	Purpose	Role
POST	/api/v1/auth/register	Create admin account	Public
POST	/api/v1/auth/login	Authenticate and issue JWT	Public
POST	/api/v1/auth/logout	Logout endpoint	Admin/Employee
GET	/api/v1/admin/employees	List org employees	Admin
POST	/api/v1/admin/employees	Create employee in same org	Admin
GET	/api/v1/admin/connections	List org connections	Admin
POST	/api/v1/admin/connections	Create connection	Admin
PUT	/api/v1/admin/connections/{id}	Update org connection	Admin
DELETE	/api/v1/admin/connections/{id}	Delete org connection	Admin
POST	/api/v1/admin/permissions	Assign employee permissions	Admin
GET	/api/v1/admin/metadata	Read org schema catalog	Admin
POST	/api/v1/admin/analyse	Run admin analysis	Admin
GET	/api/v1/employee/connections	List assigned connections	Employee
GET	/api/v1/employee/schema	List assigned schema	Employee
POST	/api/v1/employee/analyse	Run employee analysis	Employee
POST	/api/v1/employee/analyse/export/{format}	Export employee analysis	Employee

4. Internal Data Model

Collection	Key Fields	Description
users	organisation, email, role	Admins and employees with organisation scope
connections	connection_id, organisation, db_type	External datasource registry
permissions	employee_id, connection_id, can_*	Per-employee access matrix
metadata_catalog	connection_id, table_name, column_name	Discovered schema catalog
query_logs	employee_id, organisation, prompt	Execution logs for audit
counters	name, seq	Sequence IDs for entities

5. File-by-File Technical Guide

File	Responsibility
backend/app/main.py	FastAPI boot, middleware, router setup
backend/app/api/v1/auth.py	Login/register/logout and token creation
backend/app/api/v1/admin.py	Admin domain: employees, connections, permissions, metadata, analysis
backend/app/api/v1/employee.py	Employee domain: assigned connections, schema, analysis/export
backend/app/services/analysis_service.py	Cross-source analysis orchestration
backend/app/services/query_service.py	SQL prompt execution with safety checks
backend/app/services/connection_service.py	SQL/Mongo connection construction + validation
backend/app/services/metadata_service.py	Schema extraction into metadata catalog
backend/app/core/security.py	Password hash/verify and JWT encode
backend/app/utils/password_validator.py	Password strength policy on backend
frontend/app/page.tsx	Login page
frontend/app/signup-admin/page.tsx	Admin signup with strength meter
frontend/app/admin/page.tsx	Admin console UI
frontend/app/employee/page.tsx	Employee workspace UI
frontend/lib/api.ts	Frontend API client and error normalization
frontend/lib/passwordValidator.ts	Client-side password strength checks
frontend/components/ResultChart.tsx	Visualization component
frontend/components/PasswordStrengthMeter.tsx	Password strength UI component
frontend/app/api/proxy/[...path]/route.ts	API proxy with backend fallback

6. Connection and Security Notes

- Supports form mode and URL mode for datasource onboarding.
- Secrets are encrypted before storage and decrypted only during connection operations.
- Password strength validation is enforced in frontend and backend.
- JWT now carries organisation claim so UI can display org context.
- Legacy records without organisation can be backfilled for visibility after tenant isolation rollout.

End of detailed guide.